CSTE4103 — Cloud Computing

Types of Clouds, Providers, and Data Centers

Private, Public, Hybrid, Multi-Cloud, Lock-in, Hyperscale, and Data Center Essentials

Lecture 2 • Cybersecurity Engineering — 4th Year

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LECTURE 2 • OVERVIEW

Learning Objectives

1. **Deployment models**
   Define private, public, hybrid, and multi-cloud deployments.
2. **Private vs public**
   Compare them by ownership, cost, control, and security responsibility.
3. **Provider lock-in**
   Explain what causes lock-in and how to reduce it.
4. **Hybrid vs multi-cloud**
   Distinguish the two and know when each one fits.
5. **Hyperscale and data centers**
   Describe hyperscale and the building blocks of a data center.
6. **Security lens**
   Choose a deployment model by weighing risk and compliance.

1 — FRAMING

Two Questions About Any Cloud

Lecture 1 answered the first question. Today we answer the second.

Service model

WHAT do you get from the provider?

IaaS • PaaS • SaaS • DaaS
(Lecture 1)

Deployment model

WHO owns the infrastructure, and WHO shares it?

Private • Public • Hybrid • Multi-cloud
(Today)

Example: Gmail is SaaS (service model) delivered from a public cloud (deployment model). The two choices are independent.

2 — DEPLOYMENT MODELS

Private Cloud

Cloud infrastructure dedicated to a single organization, run in its own data center or in a provider's dedicated space.

Characteristics

- One tenant: no sharing with other customers
- Full control of hardware, network, and policy
- High upfront cost (CapEx) and in-house staff
- Elasticity limited by the hardware you own

Typical technologies

- OpenStack
- VMware vSphere
- Proxmox (our Labs 7–8)
- Kubernetes on your own servers

⚠ Security angle: You own the whole stack, so you also own the whole security burden: physical access, patching, monitoring, and incident response.

2 — DEPLOYMENT MODELS

Public Cloud

Infrastructure owned and operated by a provider and rented to many customers (tenants) over the internet.

Characteristics

- Multi-tenant: resources shared and isolated logically
- Pay-as-you-go (OpEx), no upfront hardware
- Near-instant elasticity and global regions
- The provider secures the infrastructure layers

Major providers

Amazon Web Services

Microsoft Azure

Google Cloud

Also: Alibaba Cloud, Oracle Cloud, IBM Cloud, DigitalOcean

⚠ Security angle: Shared responsibility: the provider secures the cloud, you secure what you put in it. Misconfiguration is a leading cause of public-cloud breaches.

2 — DEPLOYMENT MODELS

Private vs Public at a Glance

Aspect	Private cloud	Public cloud
Ownership	Single organization	Provider; shared by many tenants
Cost model	High upfront (CapEx)	Pay-as-you-go (OpEx)
Elasticity	Limited by owned hardware	Near-unlimited, on demand
Control	Full	Limited to what the provider exposes
Time to deploy	Weeks to months	Minutes
Security duty	Entirely the organization	Shared with the provider
Best for	Strict compliance, steady workloads	Variable demand, fast growth

2 — DEPLOYMENT MODELS

Hybrid Cloud

Private cloud / on-premises

Sensitive data
Legacy systems
Regulated workloads

VPN / dedicated link

⇄

Public cloud

Burst capacity
Web front-ends
Analytics and backups

Cloud bursting

Run normally in-house; rent public capacity only for peaks.

Data residency

Keep regulated data local; use the public cloud for everything else.

Gradual migration

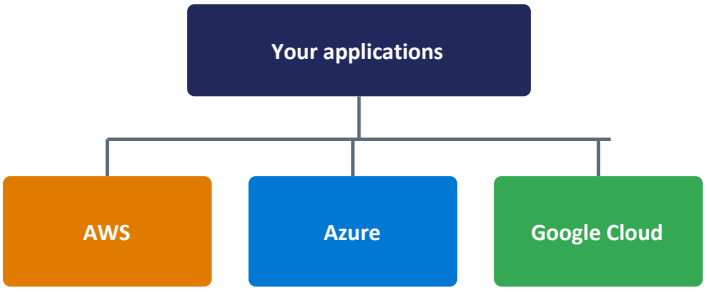
Move workloads to the cloud step by step, not all at once.

⚠ Security angle: The link between the two worlds is a new attack surface; encrypt it and keep policies consistent on both sides.

2 — DEPLOYMENT MODELS

Multi-Cloud

Using two or more public cloud providers for different workloads, or for the same workload.



Benefits

- Less lock-in
- Resilience to one provider's outage
- Best service for each task

Challenges

- More skills needed
- Different IAM and tools
- Egress fees between clouds

Hybrid is not the same as multi-cloud

Hybrid = private + public, working together. Multi-cloud = two or more public providers. A company can be both at the same time.

3 — PROVIDERS

Provider Lock-in

Lock-in means switching providers becomes so costly or difficult that you stay even when you would rather leave.

What creates it

- Proprietary services and APIs (managed databases, serverless)
- Data gravity: large datasets are slow and costly to move out
- Egress (data-out) fees
- Team skills and tooling tied to one platform
- Long-term contracts and commitments

How to reduce it

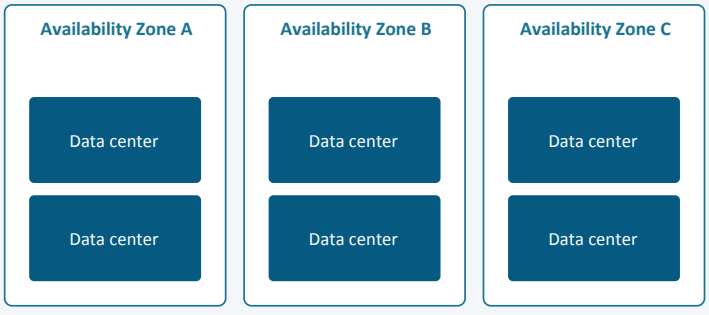
- Containers and Kubernetes for portable workloads
- Infrastructure as Code (e.g., Terraform)
- Open standards and portable data formats
- Regular exports and tested backups
- A written exit plan before you sign

⚠ Security angle: Lock-in is also an availability and resilience risk: if the provider has an outage or changes its terms, you have no quick alternative.

3 — PROVIDERS

Hyperscale

Region



Zones are separate facilities in one region, linked by fast private networks, so one failure does not take everything down.

What makes it hyperscale

- Data centers with very large server counts
- Global footprint of regions and zones
- Custom hardware, chips, and networking
- Automation of almost everything

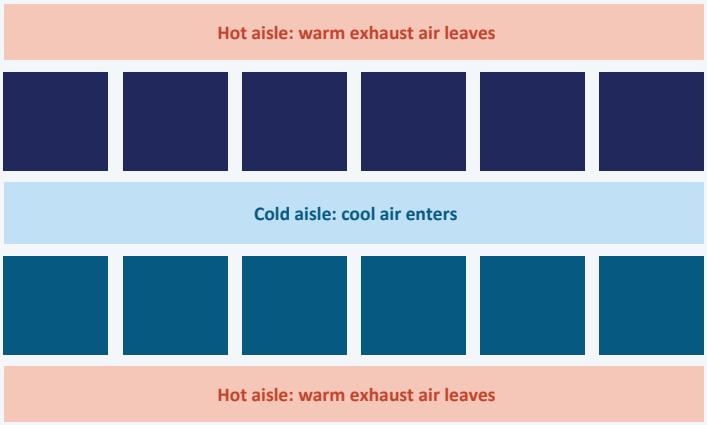
Why it matters

Economies of scale lower unit prices and allow services no single company could build alone.

⚠ Security angle: Concentration risk: a large share of the internet depends on a few hyperscalers, so one failure can ripple widely.

4 — INSIDE THE DATA CENTER

Racks, Aisles, and Pods



Rows face each other so cold air is drawn in from the middle and hot air is collected at the outer aisles.

Rack

A standard frame (about 42U) holding servers and a top-of-rack switch.

Aisle

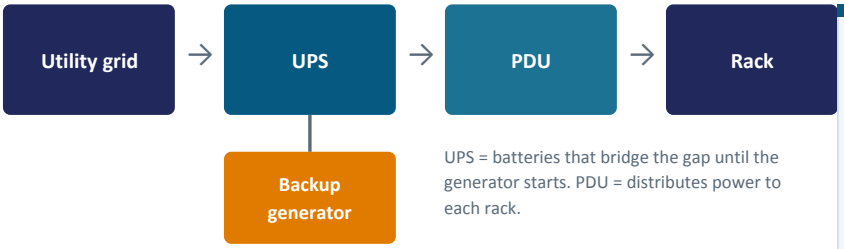
The space between rows, used to control the flow of cool and hot air.

Pod

A group of racks sharing power, cooling, and network uplinks: the unit you add as you grow.

4 — INSIDE THE DATA CENTER

Power and Cooling



UPS = batteries that bridge the gap until the generator starts. PDU = distributes power to each rack.

PUE

PUE = total facility power ÷ power used by IT equipment. 1.0 is the ideal; the lower the value, the less energy is wasted on cooling and power losses.

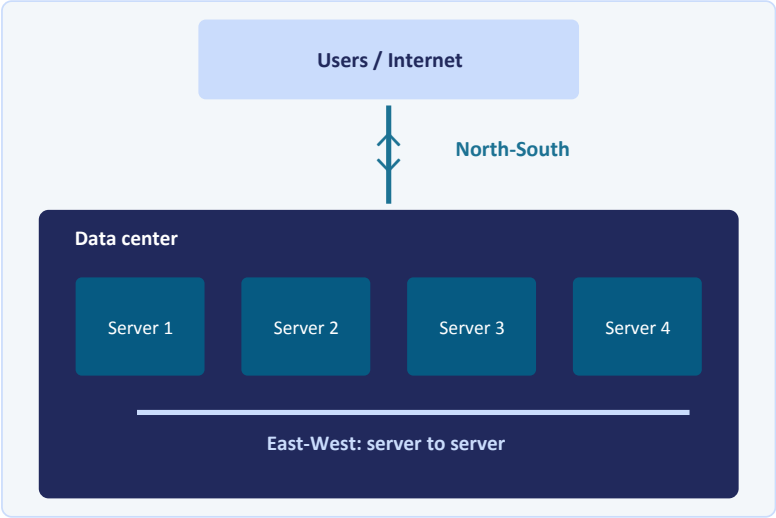
Cooling

- Hot/cold aisle containment keeps air streams apart
- Air-conditioning units (CRAC/CRAH) move the heat out
- Chilled water and liquid cooling for very dense racks
- Site and climate choices can cut cooling cost

⚠ Security angle: Availability is a security property: redundant power and cooling paths protect against outages and sabotage.

4 — INSIDE THE DATA CENTER

North-South vs East-West Traffic



North-South

Traffic between users outside and servers inside the data center.

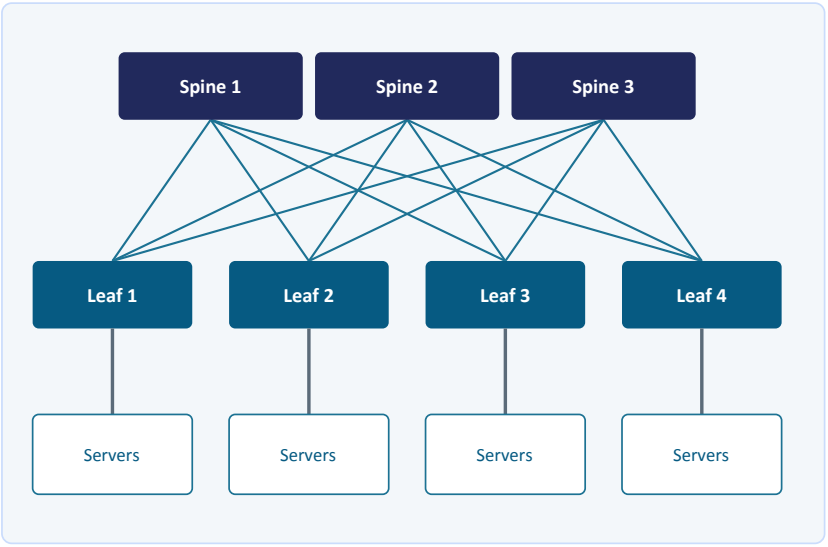
East-West

Traffic between servers inside: storage replication, microservice calls, data processing. It is usually the larger share in cloud data centers.

⚠ Security angle: Perimeter firewalls rarely see east-west traffic, so attackers move laterally. Micro-segmentation and zero trust limit it.

4 — INSIDE THE DATA CENTER

Fat-Tree and Leaf-Spine Design



Why leaf-spine

- Every leaf connects to every spine
- Any two servers are the same hops apart: predictable latency
- More servers: add leaves. More bandwidth: add spines
- Many equal-cost paths, so one failed link barely matters

Fat-Tree

The idea of many parallel equal paths with full bandwidth. Leaf-spine is a common practical form of it.

4 — INSIDE THE DATA CENTER

Unified Network and Security Takeaways

Unified data center network

One Ethernet-based fabric carries user, storage, and management traffic instead of a separate network for each.

- Fewer devices and cables
- Simpler operations
- Needs priorities and isolation so traffic types do not interfere

Security takeaways

- Control physical access to racks and cages
- Separate the management network from user and storage traffic
- Watch east-west traffic, not only the perimeter
- Treat power and cooling redundancy as availability controls

Next: all this shared hardware is sliced up by virtualization. That is Lecture 3.

5 — SECURITY LENS

Security Across Deployment Models

Model	Who secures what	Key risk	Typical controls
Private	The organization, all layers	Under-resourced patching and monitoring	Hardening, SIEM, physical access control
Public	Shared with the provider	Misconfigured storage and IAM	Least privilege, MFA, encryption, posture checks
Hybrid	Both sides, plus the link	Weak link, inconsistent policies	Encrypted tunnels, unified identity, segmentation
Multi-cloud	Each provider separately	Bigger attack surface, tool sprawl	Central identity, policy-as-code, unified logging

Rule of thumb: the more places your data lives, the more consistent your identity and logging must be.

6 — APPLY IT

Choosing a Model: Mini Case

The scenario

A national bank must keep customer records inside the country. Its mobile app has unpredictable peaks on salary days and holidays, and the board worries about depending on one vendor.

Which deployment model would you recommend, and why?

1. **Is the data regulated?**
   Points toward private or hybrid.
2. **Is demand predictable?**
   Unpredictable favors public elasticity.
3. **Do we have staff to run it?**
   No staff makes private harder.
4. **Can we depend on one vendor?**
   No points toward multi-cloud.

SUMMARY

Summary and Next Week

Key points

- Private = control and responsibility; public = elasticity and shared responsibility.
- Hybrid mixes private and public; multi-cloud mixes several public providers.
- Lock-in grows with proprietary services and data gravity; plan your exit early.
- Hyperscalers deliver scale and low cost but concentrate risk; racks, pods, power, cooling, and leaf-spine networks make it work.
- Pick the model by data sensitivity, demand pattern, skills, and risk.

Next week (Lecture 3)

Virtual Machines and Containers: approaches to virtualization, hypervisors, VM migration, Docker containers and their components.

Reference: Comer, D. E. (2021). The Cloud Computing Book. CRC Press.